



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 11 Environmental Pathology

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Learning Objectives

- ☐ Understand how the body processes and clears environmental chemicals.
- ☐ Identify the main health risks from air pollution and radiation.
- ☐ Recognize the damage caused by heavy metals like lead and mercury.
- ☐ Describe how tobacco, alcohol, and poor diet cause chronic disease.
- ☐ Compare how extreme heat, cold, and radiation injure human tissue.

Introduction

Environmental pathology is the study of how **physical, chemical, and biological agents** in our surroundings—whether at home, in the workplace, or in the general environment—cause tissue injury and disease.

Principles of Toxicology

To understand how environmental factors cause disease, we must first understand how toxins interact with the human body.

Dosage and Potency

The fundamental rule of toxicology, famously stated by Paracelsus, is:
"The dose makes the poison."

- **Acute Toxicity:** Results from a single, high-dose exposure (e.g., carbon monoxide poisoning).
- **Chronic Toxicity:** Results from repeated, low-dose exposures over time (e.g., lead poisoning or asbestos).

Metabolism of Xenobiotics

Xenobiotics are exogenous chemicals in the environment (drugs, pollutants) that may be absorbed through inhalation, ingestion, or skin contact.

The body attempts to detoxify these via two phases, primarily in the liver:

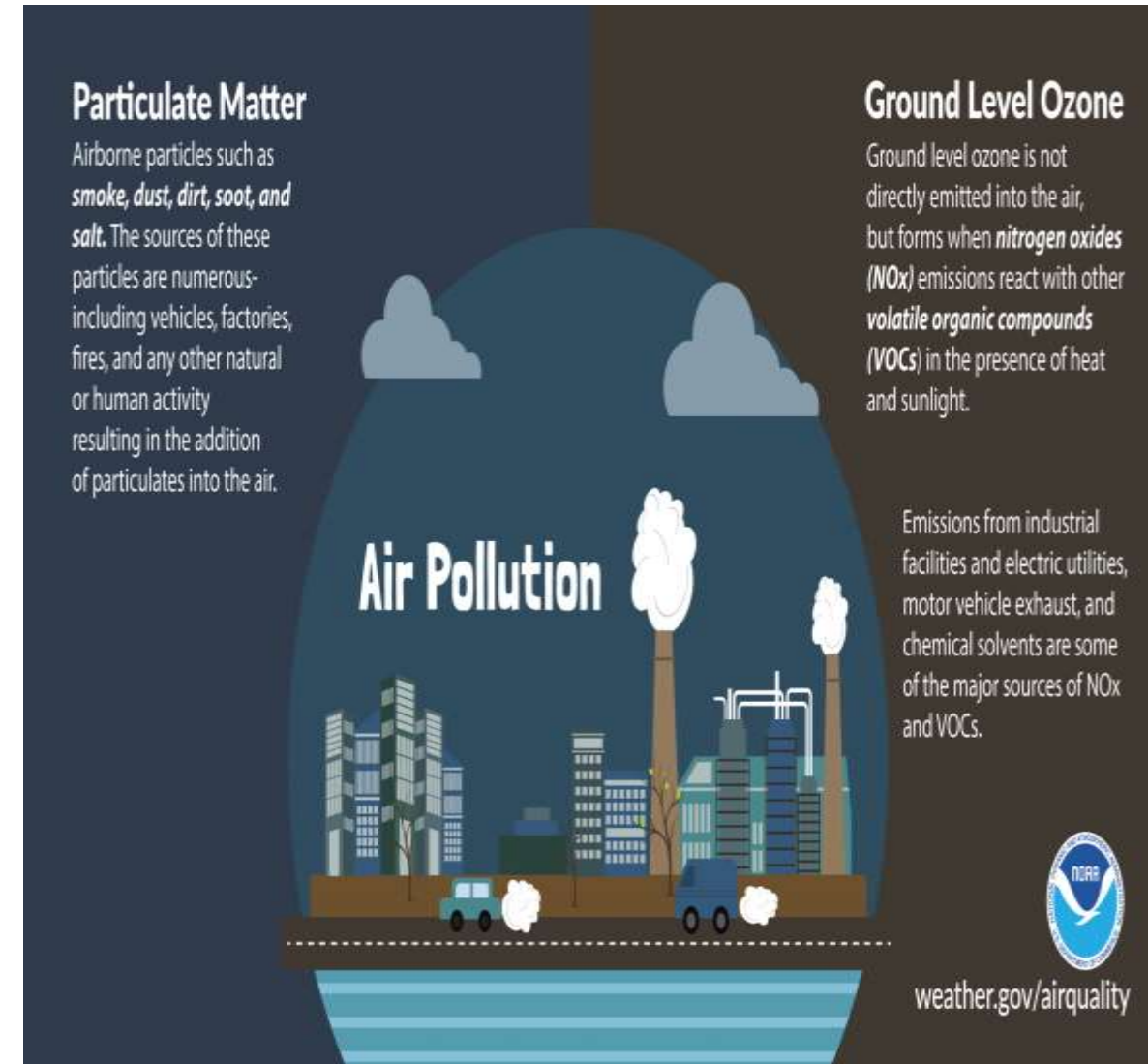
1.Phase I Reactions: Hydrolysis, oxidation, or reduction. The most important system here is the **Cytochrome P-450** enzyme system.

2.Phase II Reactions: Conjugation (solubilization) to make the toxin water-soluble for excretion in urine or bile.

Air Pollution

Outdoor Air Pollution

- **Ozone (O₃):** Formed by sunlight reacting with nitrogen oxides. It generates free radicals that damage the alveolar lining.
- **Particulate Matter (PM):** Specifically PM_{2.5} (particles < 2.5 micrometers). These are dangerous because they are small enough to reach the deep alveoli and cross into the bloodstream, triggering systemic inflammation.



Air Pollution

Indoor Air Pollution

- **Carbon Monoxide (CO):** An odorless, colorless gas. It has a much higher affinity for hemoglobin than oxygen, forming **carboxyhemoglobin**.

- *Clinical Sign:* "Cherry-red" discoloration of skin and mucous membranes.

- **Radon:** A radioactive gas derived from uranium in soil. It is the second leading cause of lung cancer in non-smokers.

Signs of carbon monoxide poisoning



Heavy Metals and Industrial Toxins

Heavy metals interfere with essential enzymatic functions,
Formation of free radicals

Toxin

Common Source

Lead

Old paint, contaminated water

Mercury

Contaminated fish (methylmercury)

Arsenic

Ground water, pesticides

Cadmium

Batteries, fertilizers



Personal Habits: Tobacco and Alcohol

Tobacco Smoke

Tobacco contains over 4,000 substances, many of which are carcinogenic

Cardiovascular: Increases atherosclerosis and myocardial infarction risk.

•**Respiratory:** Causes emphysema, chronic bronchitis, and lung cancer.

•**Maternal/Fetal:** Increases risk of spontaneous abortion and low birth weight.



Personal Habits: Tobacco and Alcohol

Alcohol (Ethanol)

Alcohol is metabolized in the liver by **alcohol dehydrogenase** into **acetaldehyde**, which is highly toxic.

- **Acute effects:** CNS depression and hepatic steatosis (fatty liver).
- **Chronic effects:** Cirrhosis of the liver, dilated cardiomyopathy, and increased risk of cancers (oral cavity, esophagus, and breast).



Physical Agents

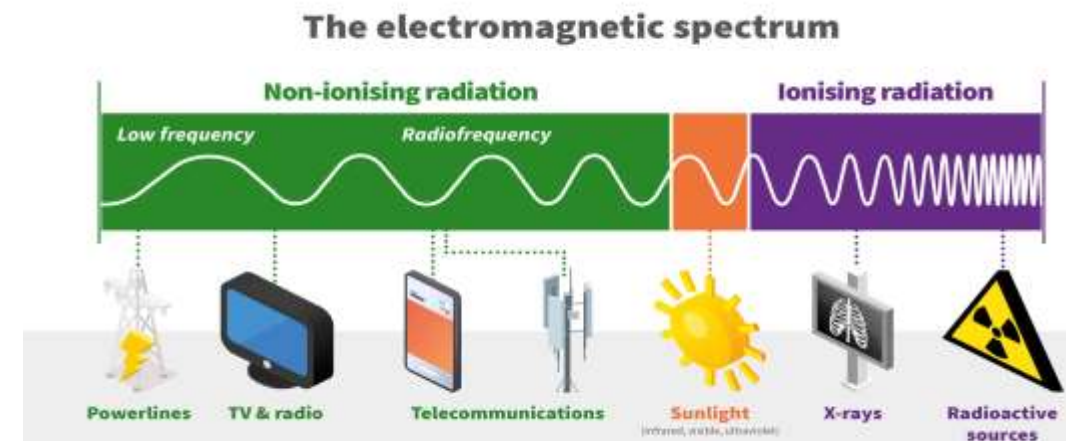
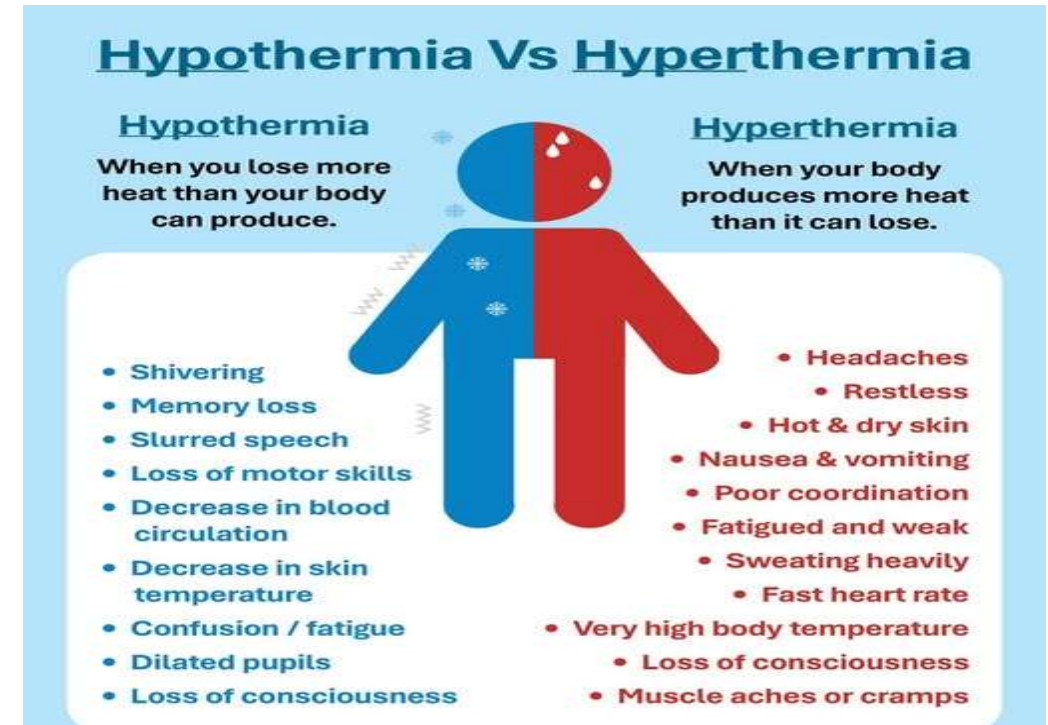
A-Thermal Injury

- **Hyperthermia:** Heat cramps, heat exhaustion, or **Heat Stroke** (failure of thermoregulation, body temp $>104^{\circ}\text{F}$ or 40°C).
- **Hypothermia:** Causes crystallization of intra- and extracellular water, leading to cell death and "trench foot" or frostbite.

B-Ionizing Radiation

Radiation damages cells either directly (DNA breaks) or indirectly by creating **reactive oxygen species (ROS)** from water.

- **High-sensitivity cells:** Hematopoietic cells, germ cells, and intestinal epithelium.
- **Long-term risks:** Leukemia and various solid tumors (thyroid, breast, lung).

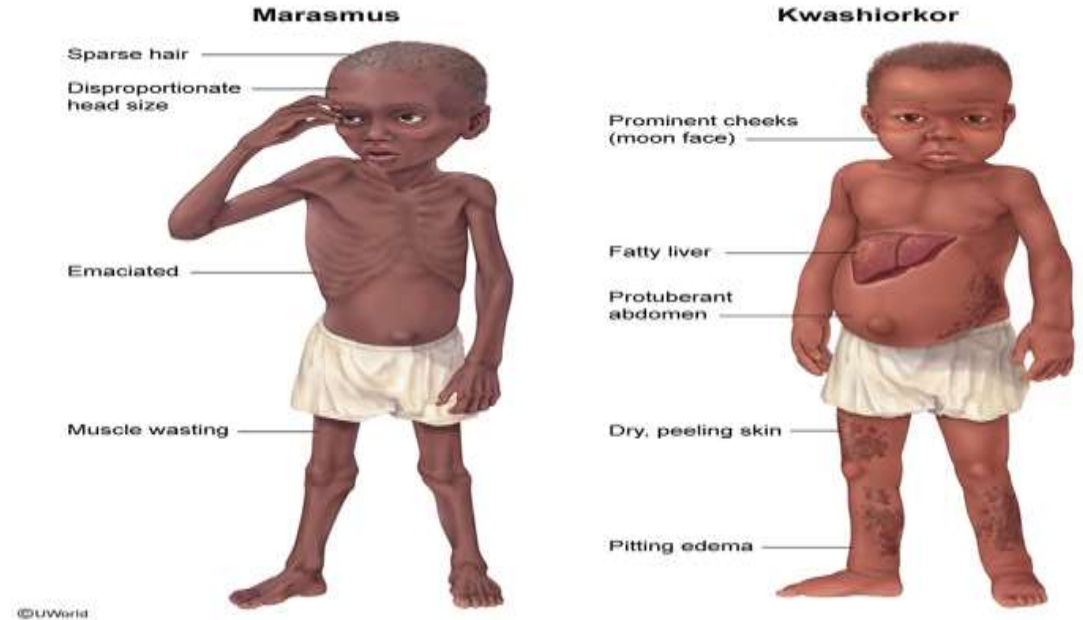


Nutritional Pathology

1-Malnutrition:

Marasmus: Calorie deficiency (weight loss, muscle wasting).

Kwashiorkor: Protein deficiency (edema, "flaky paint" skin, enlarged fatty liver).



2-Obesity: Associated with a pro-inflammatory state, increasing the risk for Type 2 Diabetes, hypertension, and certain cancers.



Nutritional Pathology

•Vitamin Deficiencies:

Vitamin A: Night blindness and immune deficiency.

Vitamin C: Scurvy (impaired collagen synthesis, bleeding gums).

Vitamin D: Rickets (children) or Osteomalacia (adults).

